

## SAN FRANCISCO BUOY 46026, CA, USA

WMO: 992640

Lat: 37.754N

Lon: 122.839W

Elev: 0

StdP: 14.70

Time Zone: -8.00 (NAP)

Period: 94-18

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 4             | 45.7       | 46.9 | N/A                           | N/A | N/A  | N/A | N/A | N/A  | 36.7                  | 49.6 | 34.5 | 49.5 | 14.0                  | 320  | 0.669 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |     |      |     |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|-----|------|-----|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%  |      | 2%  |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB  | MCDB | WB  | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k) | (l)  | (m) | (n)  | (o)                  | (p)  |
| 9             | 3.2                    | 63.8            | N/A  | 62.4 | N/A  | 61.4 | N/A  | N/A                 | N/A  | N/A | N/A  | N/A | N/A  | 9.3                  | 300  |

| Dehumidification DP/MCDB and HR |     |      |     |     |      |     |     |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-----|------|-----|-----|------|-----|-----|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |     |      | 1%  |     |      | 2%  |     |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR  | MCDB | DP  | HR  | MCDB | DP  | HR  | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b) | (c)  | (d) | (e) | (f)  | (g) | (h) | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| N/A                             | N/A | N/A  | N/A | N/A | N/A  | N/A | N/A | N/A  | N/A           | N/A  | N/A  | N/A  | N/A  | N/A  | N/A               |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |  |  |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|--|--|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |  |  |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |  |  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |  |  |
| 32.3              | 29.4 | 24.9 | 43.8                       | 67.5 | 2.4                | 2.2 | 42.1                                               | 69.1 | 40.7       | 70.4 | 39.4       | 71.6 | 37.7       | 73.2 |  |  |
|                   |      |      | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |      | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 53.8   | 52.3  | 52.3  | 51.9  | 51.5  | 52.0  | 53.1  | 55.0  | 56.6  | 57.0  | 56.3  | 54.8  | 52.7 |
|                                                                               | DBStd       | 3.15   | 2.59  | 2.64  | 2.52  | 2.01  | 2.22  | 1.93  | 2.47  | 2.73  | 2.60  | 2.67  | 2.49  | 2.76 |
|                                                                               | HDD50       | 41     | 8     | 7     | 9     | 7     | 4     | 0     | 0     | 0     | 0     | 0     | 0     | 6    |
|                                                                               | HDD65       | 4086   | 393   | 355   | 407   | 404   | 403   | 358   | 309   | 261   | 241   | 268   | 307   | 380  |
|                                                                               | CDD50       | 1432   | 80    | 73    | 67    | 54    | 66    | 92    | 156   | 204   | 209   | 197   | 143   | 91   |
|                                                                               | CDD65       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                               | CDH74       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                               | CDH80       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
| Wind                                                                          | WSAvg       | 12.7   | 12.8  | 13.9  | 14.0  | 14.5  | 14.8  | 14.1  | 11.1  | 10.8  | 10.3  | 11.1  | 12.0  | 13.4 |
| Precipitation                                                                 | PrecAvg     | 31.3   | 6.4   | 6.4   | 3.4   | 1.9   | 1.2   | 0.2   | 0.0   | 0.1   | 0.1   | 1.5   | 3.2   | 7.0  |
|                                                                               | PrecMax     | 48.8   | 16.7  | 18.9  | 10.9  | 6.7   | 4.1   | 1.0   | 0.0   | 0.9   | 0.4   | 5.3   | 10.0  | 18.4 |
|                                                                               | PrecMin     | 15.7   | 0.7   | 0.3   | 0.0   | 0.1   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.2   | 0.6  |
|                                                                               | PrecStd     | 8.7    | 4.6   | 4.1   | 3.3   | 1.6   | 1.2   | 0.3   | 0.0   | 0.2   | 0.1   | 1.4   | 2.8   | 5.0  |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 59.5  | 60.5  | 61.2  | 58.7  | 61.8  | 59.5  | 62.7  | 65.0  | 64.7  | 65.4  | 63.8  | 60.9 |
|                                                                               |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                               | 2%          | DB     | 57.8  | 58.8  | 57.6  | 56.2  | 58.3  | 58.3  | 61.4  | 63.5  | 63.4  | 62.6  | 60.6  | 59.4 |
|                                                                               |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                               | 5%          | DB     | 56.9  | 57.1  | 56.4  | 55.1  | 56.3  | 57.2  | 60.1  | 62.0  | 61.9  | 61.6  | 59.4  | 57.8 |
|                                                                               |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                               | 10%         | DB     | 56.0  | 55.7  | 55.3  | 54.3  | 55.1  | 56.0  | 58.9  | 60.6  | 60.5  | 60.2  | 58.4  | 56.6 |
|                                                                               |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                               |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                               | 2%          | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                               |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                               | 5%          | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                               |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                               | 10%         | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                               |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 3.8   | 3.6   | 3.3   | 2.8   | 2.6   | 2.7   | 2.9   | 3.1   | 3.2   | 3.6   | 3.4   | 3.8  |
|                                                                               |             | MCDBR  | 4.5   | 5.3   | 4.9   | 3.8   | 4.5   | 3.5   | 3.2   | 3.5   | 4.0   | 5.6   | 4.6   | 4.6  |
|                                                                               |             | MCWBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                               | 5% WB       | MCDBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                               |             | MCWBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.322  | 0.331 | 0.340 | 0.357 | 0.364 | 0.356 | 0.368 | 0.340 | 0.343 | 0.337 | 0.321 | 0.322 |      |
|                                                                               | taud        | 2.575  | 2.548 | 2.523 | 2.439 | 2.447 | 2.474 | 2.448 | 2.533 | 2.500 | 2.560 | 2.584 | 2.579 |      |
|                                                                               | Ebn at Noon | 275    | 287   | 294   | 292   | 291   | 292   | 287   | 294   | 288   | 280   | 273   | 267   |      |
|                                                                               | Edh at Noon | 24     | 27    | 31    | 35    | 36    | 35    | 36    | 32    | 31    | 27    | 23    | 22    |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 752    | 1046  | 1471  | 1918  | 2162  | 2335  | 2092  | 1856  | 1643  | 1241  | 882   | 654   |      |
|                                                                               | RadStd      | 101    | 126   | 125   | 143   | 161   | 121   | 108   | 117   | 95    | 85    | 61    | 93    |      |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|--------|---------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP       | HDD50 | HDD65 | CDD50 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)         | (i)   | (j)   | (k)   |
| (46) Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (41 neighbors) | N/A   | N/A     | -2.18  | N/A     | N/A   | N/A         | N/A   | -194  | +243  |

Nomenclature: See separate page